

REVIEW

Fetal macrocephaly: Pathophysiology, prenatal diagnosis and management

Shiri Shinar¹  | David Chitayat^{2,3} | Patrick Shannon⁴ | Susan Blaser⁵

¹Department of Obstetrics and Gynaecology, Division of Maternal Fetal Medicine, Ontario Fetal Centre, Mount Sinai Hospital, University of Toronto, Toronto, Ontario, Canada

²Department of Obstetrics and Gynecology, Prenatal Diagnosis and Medical Genetics Program, Mount Sinai Hospital, University of Toronto, Toronto, Ontario, Canada

³Division of Clinical and Metabolic Genetics, Hospital for Sick Children, University of Toronto, Toronto, Ontario, Canada

⁴Department of Pathology and Laboratory Medicine, Mount Sinai Hospital, University of Toronto, Toronto, Ontario, Canada

⁵Department of Diagnostic Imaging, Department of Medical Imaging, Hospital for Sick Children, University of Toronto, Toronto, Ontario, Canada

Correspondence

Shiri Shinar, Department of Obstetrics and Gynaecology, Division of Maternal Fetal Medicine, Ontario Fetal Centre, Mount Sinai Hospital, University of Toronto, Ontario Power Generator, 700 University Avenue, Toronto M5G 1X6, ON, Canada.
Email: shiri.shinar@sinaihealth.ca

Abstract

Macrocephaly means a large head and is defined as a head circumference (HC) above the 98th percentile or greater than +2SD above the mean for gestational age. Macrocephaly can be primary and due to increased brain tissue (megalencephaly), which in most cases is familial and benign or secondary. The latter may be due to various causes, including but not limited to communicating or non-communicating hydrocephalus, cerebral edema, focal and pericerebral increased fluid collections, thickened calvarium and brain tumors. Megalencephaly can be syndromic or non-syndromic. In the former, gyral and structural CNS anomalies are common. It is important to exercise caution when considering a diagnosis of megalencephaly due to limitations in the accuracy of HC measurement, lack of nomograms for specific populations, inconsistencies between prenatal and postnatal HC growth curves and progression over time. The degree of macrocephaly is important, with mild macrocephaly ≤ 2.5 SD carrying a good prognosis, especially when one of the parents has macrocephaly and normal development. Cases in which the patient history and/or physical exam are positive or when parental HC are normal are more worrisome and warrant a neurosonogram, fetal MRI and genetic testing to better delineate the underlying etiology and provide appropriate counseling.

Key points

What is already known about this topic?

- Fetal macrocephaly is a common diagnosis affecting 2% of pregnancies per definition. Its causes are numerous and its prognosis depends on the underlying etiology, degree of head enlargement, parental head circumferences and associated CNS and extra CNS findings.

What does this study add?

- In this comprehensive review, we summarize the primary and secondary causes of fetal macrocephaly, their genetic background, prenatal management and anticipated outcome. This review provides clinicians with the current understanding needed for appropriate patient counseling.

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1 | DEFINITIONS

Macrocephaly means large head and is defined as occipitofrontal head circumference (OFC or HC) above the 98th percentile or greater than 2 SDs above the mean for gestational age. As many fetuses whose HC is between +2 and +2.5 SD are normal, some clinicians use greater than +3 SD above the mean as macrocephaly of clinical significance. Macrocephaly can be primary and due to increased brain tissue (megalencephaly), which in most cases is familial and benign, or secondary and due to hydrocephalus, cerebral edema, focal and pericerebral increased fluid collections, thickened calvarium, and neoplasia among other causes (Figure 1).

The term "*relative macrocephaly*" is used to describe a head circumference that is $<+2$ SD from the mean but disproportionately large compared to the other fetal body measurements.¹

Megalocephaly denotes an increase in the weight and size of the brain due to true hyperplasia or overproduction of the central nervous system parenchyma.² The terms macrocephaly and megalocephaly have been used interchangeably, but do not imply the same process. Megalocephaly is one of the causes of macrocephaly. The differentiation between the two is important as diagnosis, prognosis and treatment differ³ with most cases of megalocephaly diagnosed postnatally or late in pregnancy.⁴

2 | PREVALENCE

The prevalence of macrocephaly, by definition, is approximately 2% of all pregnancies. Interestingly, in the neonatal population 0.9% are macrocephalic by the same definition, with $HC > +2SD$, with most being familial macrocephaly, with one or both parents exhibiting a large head.⁵ Most macrocephalic fetuses have normal postnatal growth and development with the HC being between $+2SD$ and $+2.5SD$ with a higher prevalence in males.⁵ It should be noted that

the sonographic diagnosis of an enlarged fetal HC has low specificity with a high number of false positive cases. Indeed, only 67% of the children diagnosed with fetal macrocephaly are born with a truly enlarged HC. Furthermore, at the time of the neuropsychological evaluation in childhood, only 35% remain macrocephalic.

The prevalence of megalocephaly, a subset of macrocephaly, at birth is unknown because imaging to exclude other causes of macrocephaly may not be made prior to birth and many of the differentiating clinical features may not be identified until birth or following genetic evaluation. Assessment of fetal brain volume is not commonly performed, but the correlation can be made on both US and MRI by specific cortical mantle measurements and correlated with normative data provided by autopsy specimens.^{6,7} Fetal brain-volume measurement can be performed by fetal MRI using semi-automated post processing methods.⁸

3 | DIFFERENTIAL DIAGNOSIS

There are numerous causes for macrocephaly. Herein, we separate the causes into two main groups - megalencephalic macrocephaly (primary) and non-megalencephalic macrocephaly (secondary) (Figure 2).

Non megalencephalic macrocephaly is macrocephaly with normal encephalon dimensions. This can be a result of non-communicating hydrocephalus with cerebro-spinal fluid (CSF) flow obstruction due to aqueduct stenosis, tumor, venous congestion or intraventricular hemorrhage and scalp edema or communicating hydrocephalus due to lack of absorption caused by malfunction of the arachnoid granulation (also known as Pacchionian granulations) secondary to infection, subarachnoid hemorrhage and intraventricular hemorrhage causing adhesive arachnoiditis. Other causes of secondary macrocephaly include abnormal calvarial thickness due to skeletal osteodysplasias, such as craniometaphyseal dysplasia (ANKH) or storage

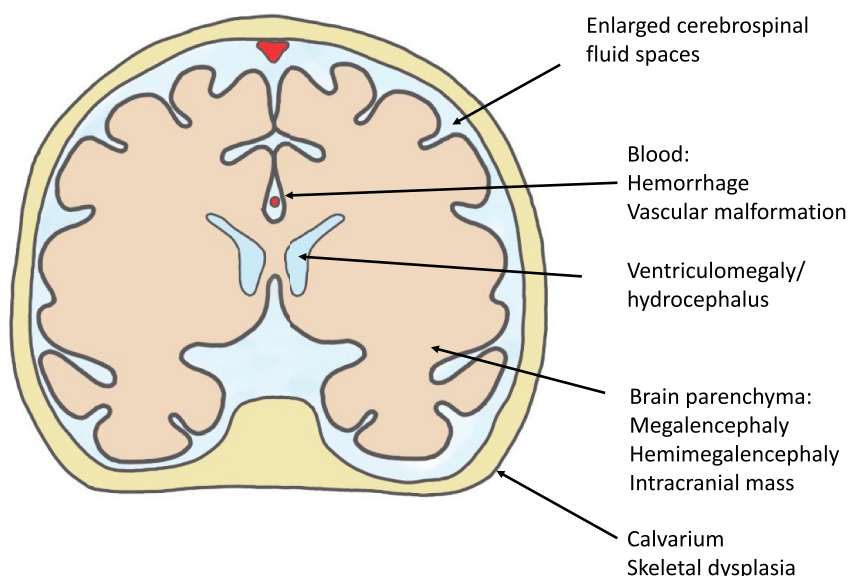


FIGURE 1 Etiologies for macrocephaly according to the cerebral compartment involved. [Colour figure can be viewed at wileyonlinelibrary.com]

disorder. In the latter, the accumulation of metabolic substances or astrocyte swelling is frequently associated with megalencephaly, however this initial brain enlargement is followed by atrophy in the disease course and the macrocephaly typically does not become apparent until later in infancy or in childhood.

In this review, we elaborate on selected causes that can be identified prenatally:

Extra-ventricular obstructive hydrocephalus - The main differential diagnosis of macrocephaly is often noted to be a transient CSF flow disturbance, also known as benign external hydrocephalus (BEH). There are various alternative terms for this disorder, including benign enlargement of the subarachnoid spaces and benign subdural hygromas or collections (Figure 3). Pericerebral fluid prominence has only rarely been described in the fetus.⁹ The most common theory

suggests that external hydrocephalus is caused by immature arachnoid villi not able to absorb the CSF that is continuously produced.¹⁰ HC is usually near normal at birth, reaching 97.5 percentile between 3 and 8 months of age. In addition to the rapidly increasing HC, there may be a tense anterior fontanel, dilated scalp veins, motor delay and irritability. The prominent fluid spaces largely resolve between 12 and 18 months of age.¹¹ The HC, however, remains enlarged, stabilizing at around the 98th percentile. Most infants are male (86.4%). A recent study on 42 children with BEH showed that nearly half have temporary or permanent psychomotor delays, suggesting that the condition is not benign as the name implies.¹²

Ventriculomegaly with or without hydrocephalus - The incidence of ventriculomegaly is 0.3 to 2.0 per 1000 pregnancies.^{13,14} The diagnosis of fetal ventriculomegaly is typically made on prenatal

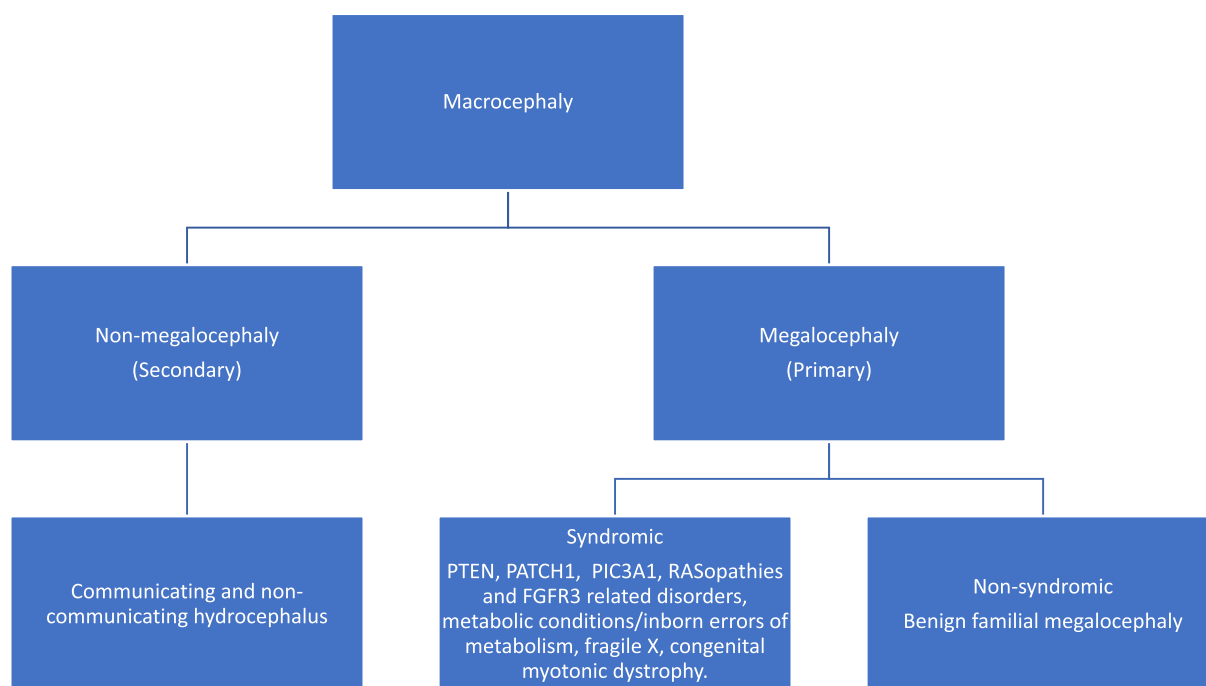


FIGURE 2 Differential diagnosis of the pathophysiology of fetal macrocephaly. [Colour figure can be viewed at wileyonlinelibrary.com]

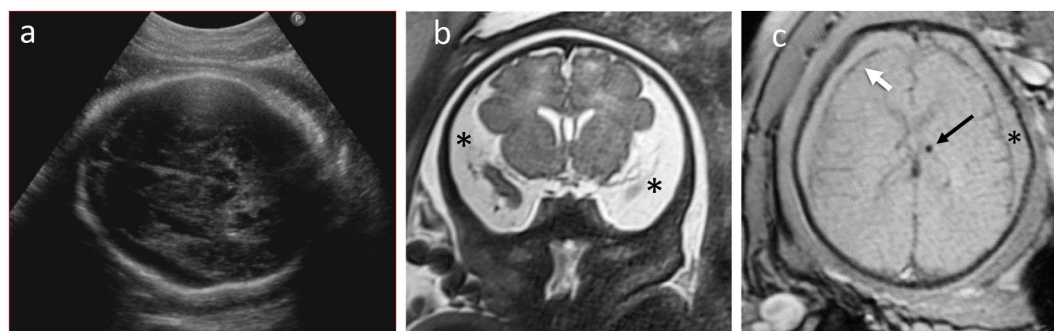


FIGURE 3 Macrocephaly secondary to benign external hydrocephalus. Axial US image (A) at 32 weeks demonstrates benign subdural hygromas. Coronal T2 MRI image (B) at 37 + 0 weeks with HC corresponding to 40 + 2 weeks confirms prominent pericerebral fluid (*) with membranes. Axial SWI (C) shows tiny focus of hemosiderin (black arrow) in left germinal matrix and lining (white arrow) the hygromas (*). [Colour figure can be viewed at wileyonlinelibrary.com]

ultrasound and is defined by an atrial diameter of the lateral ventricle that is ≥ 10 mm. When the ventriculomegaly produces increased intracranial pressure (ICP), the condition is termed hydrocephalus. In the absence of increased ICP, the ventriculomegaly is called *ex-vacuo*, that is, it results from the loss of brain volume. This type of ventriculomegaly is not associated with macrocephaly.

Although the definitions of mild and moderate ventriculomegaly differ depending on the classification system,^{15–17} it is agreed that a ventricular width greater than 15 mm is defined as severe ventriculomegaly. The prognosis of isolated ventriculomegaly depends on the degree of ventricular dilation, its progression (defined as an increase in the ventricular measurement of more than 3 mm) and the underlying cause.

Causes of ventriculomegaly resulting in macrocephaly are numerous and can be grouped into four main groups: malformative (such as holoprosencephaly, agenesis of the corpus callosum, malformation of cortical development, Dandy-Walker Malformation), obstructive (congenital aqueduct stenosis, midbrain-hindbrain malformation, open neural tube defects, cerebral tumors) and clastic (infection, stroke/ischemia, and intra-ventricular hemorrhage).¹⁸ Obstructive ventriculomegaly can result in hydrocephalus. In this case the ventricular dilation produces elevated ICP.¹⁹ The most common cause of obstructive hydrocephalus, responsible for approximately 20% of cases, is congenital aqueduct stenosis (CAS).²⁰ Aqueduct obstruction may be complete or partial resulting in dilation of the lateral and third ventricles (Figure 4). The causes for aqueduct obstruction may be genetic, vascular (such as intra-ventricular hemorrhage (Figures 5 and 6) or infectious (cytomegalovirus, toxoplasma,

syphilis etc), and the prognosis depends on the underlying etiology. As opposed to other causes of severe ventriculomegaly, isolated CAS causes non-communicating pressure hydrocephalus but with otherwise preserved brain anatomy.

Other causes of obstructive hydrocephalus are fetal brain tumors (FBT). These are very rare, occurring in 0.34/1,000,000 live births and account for 10% of all antenatal tumors.²¹ While pediatric brain tumors are typically infra-tentorial, FBTs are often supra-tentorial, with 50% of these tumors being teratomas²¹ (Figure 7). Diagnosis is usually made late in pregnancy and prognosis, which depends on the time of detection, tumor growth, size and histology, is guarded.²² MRI is valuable in providing additional prognostic information beyond that of ultrasound.²³ Space occupying lesions can also be cerebral cysts in different locations that may have a mass effect and enlargement of the HC (Figures 8–10).

4 | MEGALOCEPHALIC MACROCEPHALY

Megalocephaly may be isolated, or associated with other syndromic, and non-syndromic conditions with some being the result of single gene disorders² (Table 1). Clinically significant megalocephaly, HC $> +2.5$ -3SD, is divided into a metabolic type and an anatomical type (developmental).²⁴

1. **Anatomic megalocephaly** – The most common cause of megalocephalic macrocephaly is Benign familial megalocephaly. In this disorder, initial imaging may demonstrate benign external

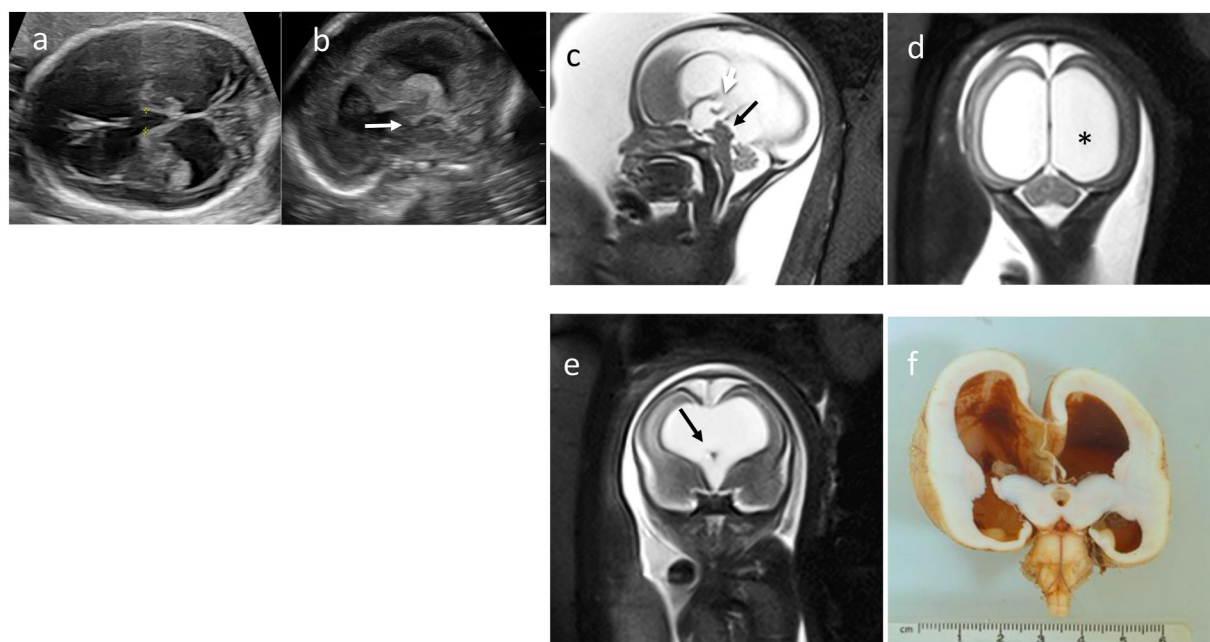


FIGURE 4 Aqueductal stenosis. US images at 22 wks demonstrate (A) triventriculomegaly and (B) aqueductal stenosis (white arrow). MR SSFSE at 22 + 5 midsagittal demonstrates (C) ventriculomegaly, caudally displaced fornical columns (white arrow) and aqueductal stenosis (black arrow). Coronal (D) confirm dilated lateral ventricles (*). Coronal (E) reveals fused fornical columns (arrow). (F) Post mortem demonstrates dilated. [Colour figure can be viewed at wileyonlinelibrary.com]

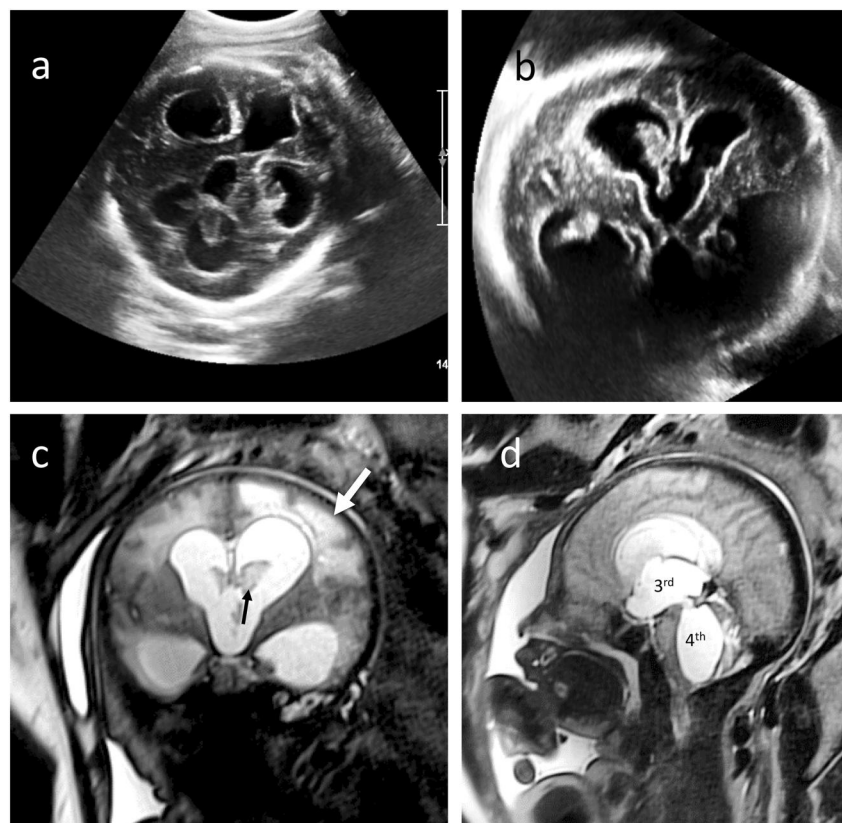


FIGURE 5 Third trimester intraventricular hemorrhage resulting in hydrocephalus. Axial (A) and coronal (B) US images at GA 35 + 5 demonstrate severe dilation of the lateral, third and fourth ventricles with visible blood clots in the ventricles. Hyperechogenicity (B) surrounding the larger ventricle. Coronal fetal MRI image (C) at GA 38 + 4 confirms the persistence of clot (black arrow) and ventriculomegaly. Hyperechogenicity on US reflects the brain edema (white arrow) shown on MRI. Sagittal image (D) shows dilatation of the 3rd ventricle, the 4th ventricle and the intervening aqueduct of Sylvius.

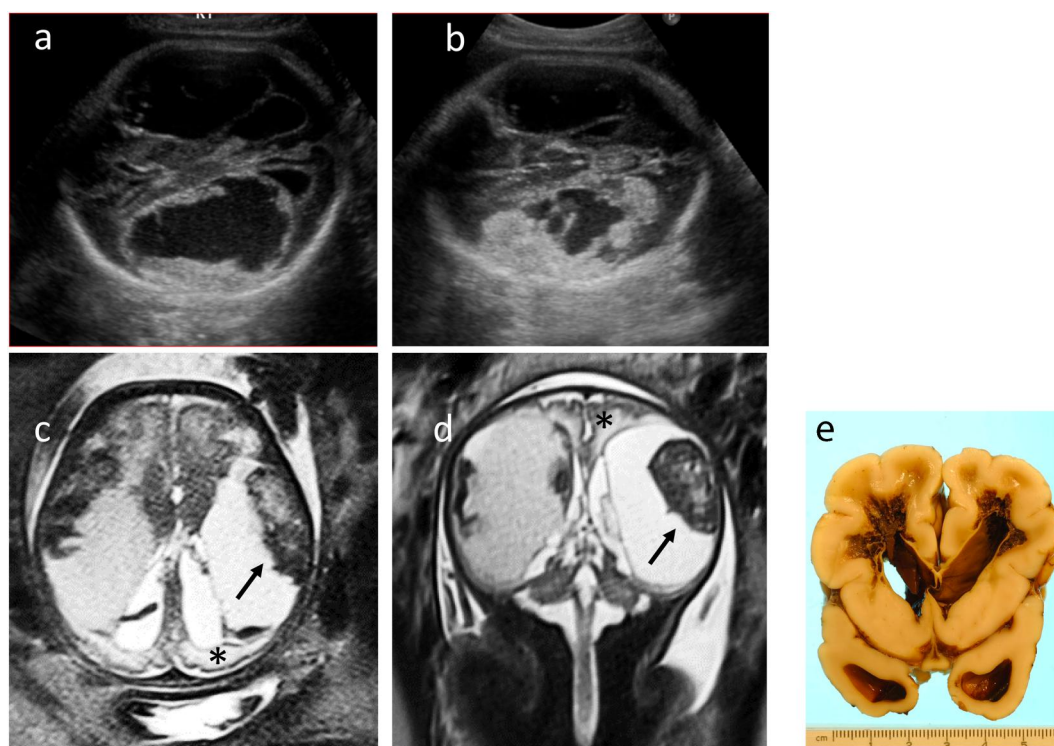


FIGURE 6 Bilateral ischemia and hemorrhage. Ultrasound axial images (A + B) at GA 30 + 5 following hemorrhagic infarct demonstrate severe bilateral ventriculomegaly with multiple blood clots along the ependymal walls and thinning of the proximal cortical mantle. Coronal (C) and axial (D) MRI images at 31 + 5 weeks with HC corresponding to 39 + 3 days confirm bilateral thrombus (arrows), layering blood, hydrocephalus and asymmetric signal of the ventricles due to blood breakdown products. The cortical mantle is markedly thinned (*). (E) 36-week brain with post hemorrhagic hydrocephalus, arising after periventricular haemorrhagic infarct with associated cystic white matter injury. [Colour figure can be viewed at wileyonlinelibrary.com]

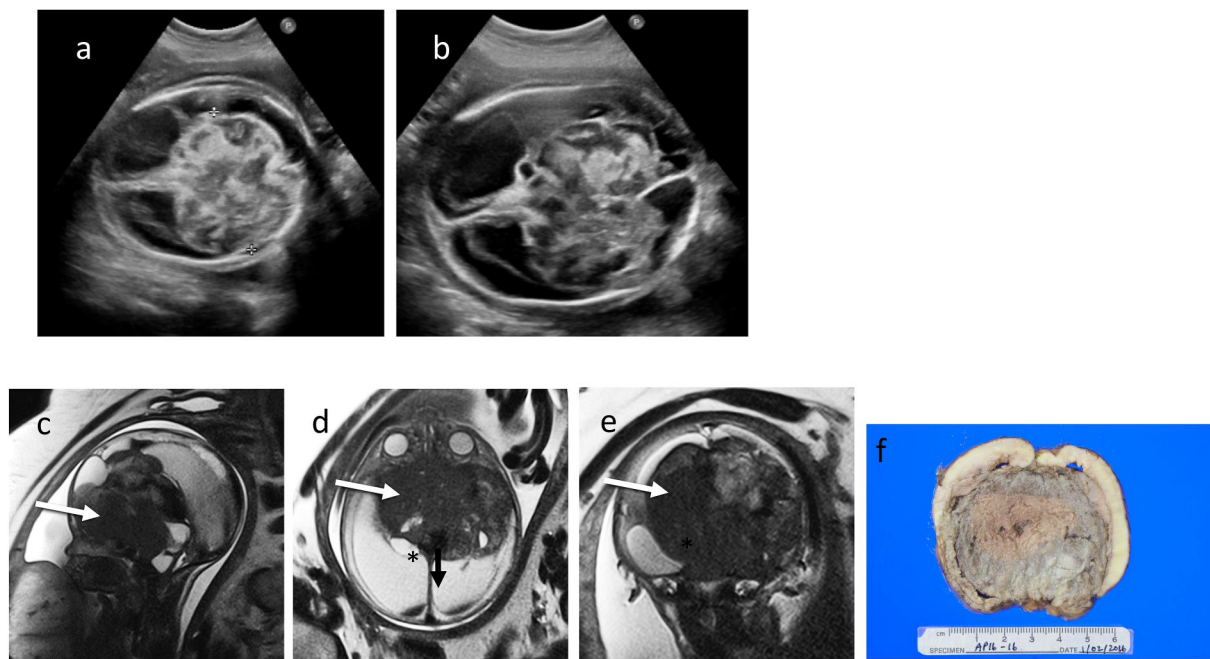


FIGURE 7 Fetal craniopharyngioma. Axial US images (A, B) at GA 30 + 1 weeks show massive heterogeneous brain tumor distorting entire brain anatomy and resulting in macrocephaly with HC of +3.47SD. Sagittal (C), axial (D) and coronal (E) T2 MRI images AT ga 30 + 2 confirm heterogenous mass (white arrows), hydrocephalus (*), thinned cortical mantle and layering intraventricular blood (black arrow). (F) Congenital craniopharyngioma at 23 weeks, replacing almost all central brain elements. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

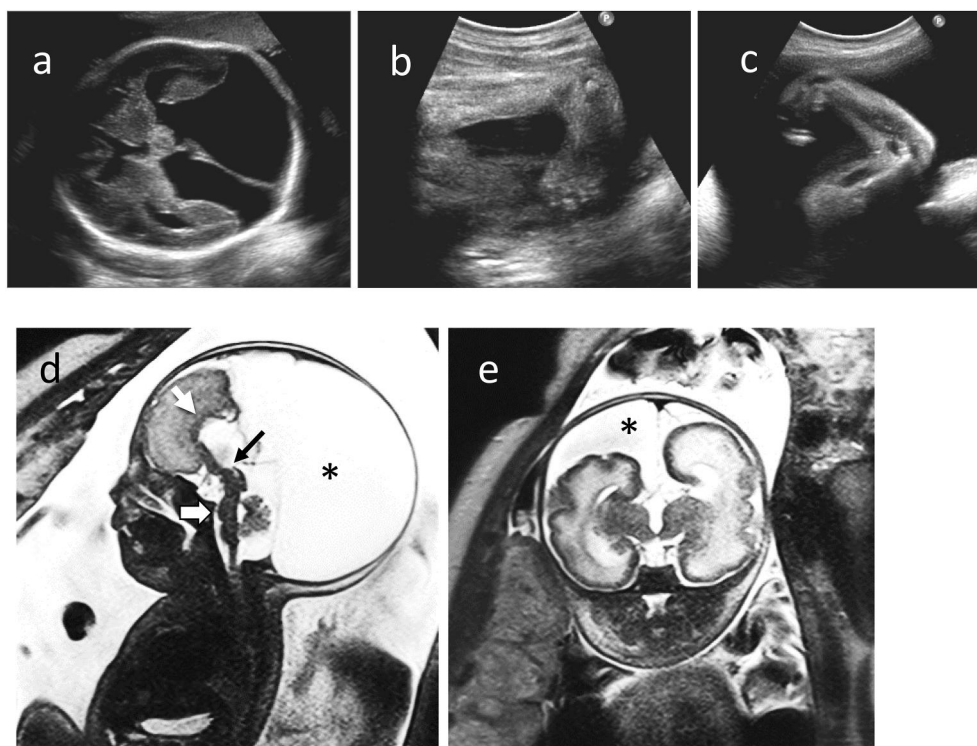


FIGURE 8 Extra axial fluid collections. Severe macrocephaly and midline distortion due to extra axial fluid collections shown on axial ultrasound (A) in a GA 27 + 5 week old fetus with multiple congenital anomalies including arthrogryposis with club feet (B) and fixed wrist flexion (C). Sagittal (D) and coronal (E) MRI images at GA 31 + 4 (HC now corresponding to GA 48 + 6 weeks) demonstrate large extra-axial collections (*), anterior remnant of the corpus callosum (white arrow), closed aqueduct (black arrow) and thin brainstem (open arrow).

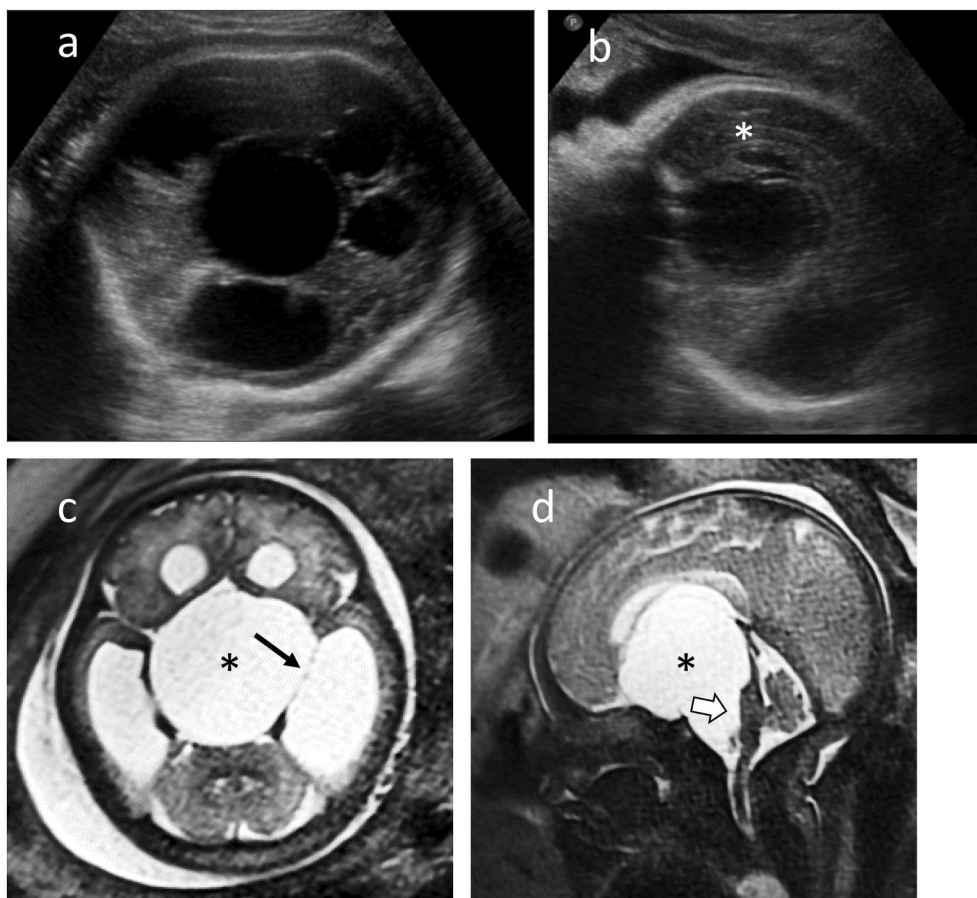


FIGURE 9 Suprasellar cyst. Axial (A) US at 34 + 0 wks shows severe bilateral ventriculomegaly, while sagittal US (B) reveals stretched and thinned corpus callosum (*). Corresponding axial MR image (C) at GA 34 + 4 weeks with HC corresponding to GA 39 + 6 weeks shows suprasellar cyst (*). Cyst wall is demonstrated (black arrow). Note the mass effect on brainstem on sagittal image (D) by the cyst (*) with severe displacement and flattening of the brainstem (open arrow).

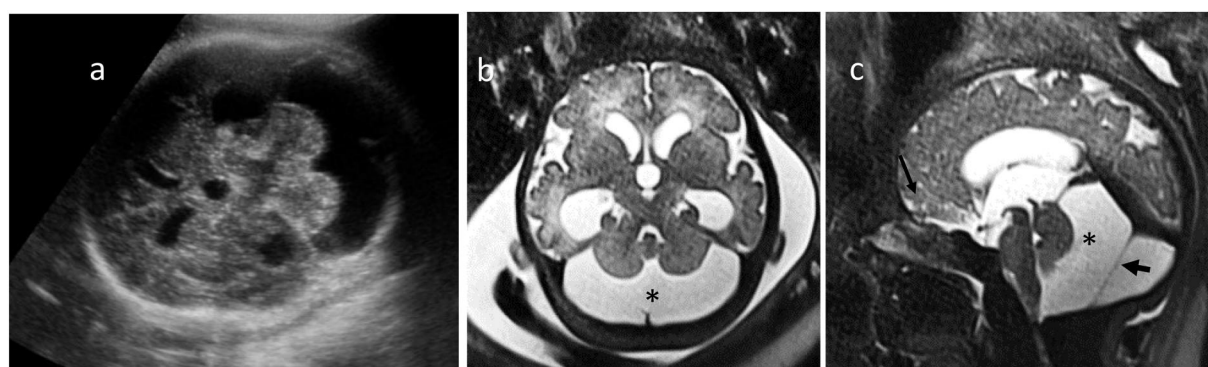


FIGURE 10 Retrocerebellar cyst. (A) Axial ultrasound at GA 37 + 5 weeks. Prominent anterior horns of the lateral ventricles with a large retrocerebellar cyst. MR images in (B) axial and (C) sagittal planes at GA 38 + 4 weeks with HC corresponding to GA 41 + 1 weeks confirm retrocerebellar cyst (*) and increasing ventriculomegaly. Persistent occipital sinus (short black arrow) distorts the straight sinus. The vermis is pushed ventrally, having a mass effect on the inferior aqueduct of Sylvius. [Colour figure can be viewed at wileyonlinelibrary.com]

hydrocephalus, but the brain remains large as the peri-cerebral spaces resolve. This condition is usually associated with a normal cognitive outcome. It is typically identified when both or one of the parents have a large head, which raises the possibility of an autosomal dominant mode of inheritance. As the

subarachnoid space normalizes, brain volume remains large, and the head circumference stabilizes usually around the 98th percentile. Most children are functionally normal. Measurement of the parental and sibling HC is important in establishing this diagnosis.

2. **Metabolic megalcephaly** – This refers to various storage and degenerative encephalopathies. As in osteodystrophies, these often become apparent later in infancy, although some can present prenatally. Examples include mucopolysaccharidoses and other lysosomal storage disorders, as well as Alexander (GFAP mutation), Canavan disease (ASPA deficiency) and megalcephalic leukoencephalopathy with subcortical cysts (MLC1).^{4,25} Metabolic storage diseases such as mucopolysaccharidoses and mucopolipidoses can also produce macrocephaly due to the deposition of abnormal metabolites in the meninges and arachnoid granulation, resulting in cerebrospinal fluid malabsorption.²⁶
3. **Other syndromic megalcephalies** – Many megalcephalic malformations result from mutations in genes that encode components of the mammalian target of rapamycin (mTOR) pathway, an intracellular signaling pathway that plays an important role in regulating cellular growth and homeostasis.²⁷ The hyperactivation of the mTOR pathway through the activation of PTEN, PIK3CA, or AKT3 results in ribosome biogenesis and elevated messenger RNA translation and, eventually, increased neuronal proliferation with production of dysmorphic neurons seen in dysplastic megalcephalies. Loss of function of the TSC1 and TSC2 genes in tuberous sclerosis also leads activate the mTOR and downstream elements (Figure 11). Klippel-Trenaunay-Weber, CLAPO and

CLOVES syndromes are PIK3CA related disorders. Other disorders in this pathway include Proteus (AKT1), Bannayan-Ruvalcaba-Riley/Cowden (PTEN), megalcephaly capillary malformation MCAP (PIK3CA) and megalcephaly, polydactyly and polymicrogyria hydrocephaly syndrome MPPH1 and 2 (PIK3R2 and AKT3, respectively), both of which have a large HC at birth.^{28–32} In addition to the disorders involving the PTEN, PI3K-PKB/Akt and mTOR pathways, megalcephaly is known to be associated with **overgrowth syndromes**. Some of these syndromes are associated with macrocephaly in combination with generalized somatic overgrowth. Examples include Sotos, Weaver, Simpson-Golabi-Behmel, Macrocephaly-Cutis Marmorata Telangiectasia Congenita (M-CMTC) and more.³³ In some of these conditions the macrocephaly is evident prenatally (Simpson-Golabi-Behmel, Sotos and Weaver) while in others only postnatally. Syndromic overgrowth conditions can also be associated with hemimegalcephaly (see below). Megalocephaly can also accompany the RASopathies **syndromes** caused by variants in genes encoding components of the RAS/MAPK signal transduction pathway. The RAS-MAPK pathway has roles in cellular proliferation, differentiation, and migration with a variable effect on brain volume and callosal size with the most common RASopathies being Noonan syndrome due to PTPN11 and Neurofibromatosis Type 1. Other RASopathies include cardiofaciocutaneous, Costello and Legius syndromes, epidermal nevus syndrome and capillary malformation-arteriovenous malformation syndrome (CM-AVM). Lastly, additional syndromes associated with megalcephaly include Weaver syndrome (EZH2), fragile X (FMR1), and congenital myotonic dystrophy³⁴ as well as chromosomal disorders such as Klinefelter syndrome (47, XXY).²

4. **Skeletal dysplasias** – Several skeletal dysplasias can produce relative and absolute macrocephaly. Mutations in the FGFR3 gene cause achondroplasia, hypochondroplasia and thanatophoric dysplasia,³⁵ all of which present with their distinct phenotype of short long bones with mainly rhizomelia related to the inhibition of endochondral bone growth.³⁶ In achondroplasia, the membranous cranial vault is disproportionately large, with associated frontal

TABLE 1 Syndromes associated with megalcephaly and hemimegalcephaly.

- Isolated (Benign familial megalcephaly)
- PTEN and PI3K-AKT-MTOR, including vascular syndromes
- RASopathies, RAS/MAPK signal transduction pathway (NF1, Costello etc.)
- FGFR3 mutations, increased brain:body ratio with relative or absolute megalcephaly (achondroplasia, thanatophoric dysplasia)
- Metabolic and degenerative disorders (mucopolysaccharidoses, other lysosomal disorders, Alexander (GFAP) and Canavan (ASPA), megalcephaly with cysts (MLC1).

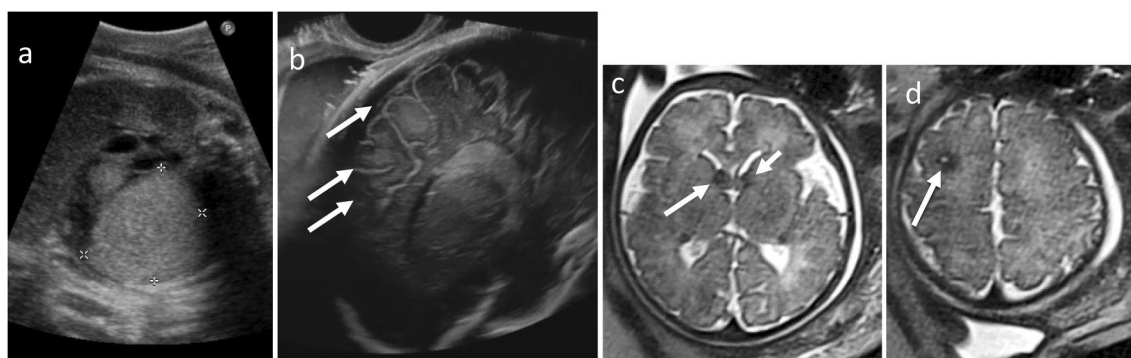


FIGURE 11 Tuberos Sclerosis. Multiple cardiac rhabdomyomas are shown on US (A) and cerebral cortical tubers (white arrows, B) at GA 34 + 0 weeks. Axial Fetal MRI (C) reveals bilateral nodules at the foramina of Monro (long arrow) and peritrial subependymal nodules (short arrow) in a GA 31 + 4 fetus with tuberous sclerosis and macrocephaly. Cortical tuber (arrow, D) is shown on axial MRI (D).

bossing and depression of the nasion.³⁷ In the neonates with achondroplasia, the head circumference is more than 2 SD above the mean and it increases even further in the first year of life. Additionally, the narrow foramen magnum impedes cerebrospinal fluid drainage, resulting occasionally in ventriculomegaly with increased head size.³⁷ Moreover, as in thanatophoric dysplasia, the brain/body weight ratio is increased; thus, macrocephaly is also a result of relative megalcephaly. Varying degrees of accompanying temporal lobe dysplasia can be observed in both skeletal dysplasias.

5. **Autism spectrum disorder (ASD)** – It has been suggested that up to 15% of individuals with ASD have macrocephaly.^{38,39} However, this is almost never a fetal finding, but only seen later in life. Cases of extreme macrocephaly (>3SD) have been correlated with mutations in the gene phosphatase and tensin homolog (PTEN).⁴⁰ These mutations cause excess neurogenesis/neuronal proliferation, suggesting this as the underlying pathomechanism for the increased cerebral size in ASD.
6. Recently, an association has been noted between increased volume of the ganglionic eminences (GE) and megalcephaly. Excessive and abnormal neuroblast proliferation and migration may explain this observed association. Similarly, unilateral enlargement of the GE is associated with ipsilateral hemimegalcephaly.⁴¹ An increased volume of the ganglionic eminence can result from various neurodevelopmental disorders, often as an indirect sign of overgrowth conditions like mTOR.

5 | HEMIMEGALOCYPHALY

Hemimegalcephaly (HME) is a rare congenital hamartomatous malformation of the brain, remarkable for its extreme asymmetry. It is characterized by enlargement and overdevelopment of one cerebral hemisphere. Many cases of HME belong to the spectrum of disorders related to hyperactivation of the mTOR pathway,⁴² which is responsible for unilateral brain and spinal cord overgrowth, the result of somatic mutations in the mTOR, *PIK3CA* or *AKT3* genes or may be a part of neurocutaneous or somatic hemihypertrophy syndromes. The former includes epidermal nevus syndrome, Proteus syndrome, Klippel-Trenaunay-Weber syndrome, neurofibromatosis type 1, and tuberculosis, among others.

6 | DIAGNOSIS

The fetal HC should always be measured during a sonographic examination performed in the second or third trimester. If the value is larger than +2 SD and/or there is disproportion with the size of the trunk or limbs, macrocephaly should be considered and the cause should be further investigated. The use of appropriate charts should be stressed since the variability between them can be very significant. This fetal diagnosis should be approached carefully due to limitations in the accuracy of HC measurement (particularly in late

gestation), lack of nomograms for specific populations based on fetal sex, ethnicity, and parental HC and inconsistencies between prenatal and postnatal HC growth curves and progression over time.⁴

Detailed history and sonographic anatomical survey should be performed to differentiate between isolated and non-isolated (syndromic) macrocephaly as well as primary and secondary macrocephaly and detect conditions associated with a relatively large head, such as intrauterine growth restriction, skeletal dysplasia, ventriculomegaly with or without hydrocephalus, fetal tumors and triploidy. Once these and other causes for macrocephaly have been determined, detailed neurosonography⁴³ is recommended.

When the HC is at the upper part of the normal range prenatally or just above +2SD, no structural brain or body abnormalities are identified, and there is a parent, sibling or other first or second degree family members with macrocephaly, or the macrocephaly can be traced through several generations, BEH should be taken into consideration as a possible diagnosis.

In syndromic megalcephaly, the HC is typically >2.5 SD and presents earlier in the pregnancy, whereas in non-syndromic megalcephaly, the large HC is detected later (early third trimester, rarely before 28 weeks, vs. mid third trimester, respectively). Many of the features that point towards a diagnosis in syndromic megalcephaly may be extracranial, as enlarging brain volume and head circumference may become apparent only after the initial anatomy scan or even after birth. These features include increased nuchal translucency/cystic hygroma and increased nuchal fold and cardiac anomalies (RASopathies), cardiac rhabdomyomas (TSC) or limb anomalies or overgrowth. While genetic mutations may affect all body cells, somatic mosaicism is common in the asymmetric somatic disorders and affected tissue biopsy may be needed to establish a diagnosis.⁴⁴

A variety of gyral and structural CNS anomalies are also common in syndromic megalcephaly. These may include mild ventriculomegaly, callosal anomalies, mostly thick corpus callosum, malformations of cortical development (polymicrogyria, pachygyria, overdeveloped sulcation, heterotopias), large CSP and large extra-axial spaces.⁴ Enlargement of the ganglionic eminence may also be seen. Assessment of the cortical mantle layering and gyral development has been documented on both fetal US and fetal MRI.⁴⁵ The cerebral findings can be similar in both isolated and syndromic forms, although tubers in TSC may be present on fetal US and MRI and extracranial features involving visceral and limb abnormalities can aid in differentiation.⁴⁶

In HME, the fetal ultrasound and brain MRI show hemispheric asymmetry with overgrowth and abnormal texture and gyration of the larger hemisphere, and often ventriculomegaly and the other hemisphere is compressed and distorted.⁴⁷ On MR, disruption of the radial glial fibers ("transient structures") of the developing hemisphere is characteristic. Asymmetric prominent diffusion-weighted imaging is present in the involved hemisphere on fetal MRI.⁴⁸

Somatic, brain stem and posterior fossa asymmetry may occur, albeit less frequently.⁴⁹ Associated CNS pathologies, such as agenesis of the corpus callosum, Dandy-Walker malformation or abnormal cerebellum may be present.

7 | INVESTIGATIONS

When the fetal HC is greater than +2 SD for gestational age, the HC of both parents and siblings should be measured and a detailed history obtained. A referral to genetics is prudent in these cases. In cases of familial large heads, a physical examination for stigmata of autosomal dominant conditions should be completed. With a negative examination and a HC <2.5 SD above the mean, follow up ultrasound measurements every 3–4 weeks are recommended. The parents should be counselled that the prognosis is usually good with normal neurodevelopmental outcomes. However, the limitations of prenatal radiological investigation and our inability to diagnose functional abnormalities should be discussed. When the history and/or physical exam are positive or when parental HC are normal, a targeted neurosonogram, a fetal MRI and an amniocentesis for chromosomal microarray (CMA) should be offered to better delineate the underlying etiology and provide appropriate counseling. Whole exome sequencing should be considered in cases of normal CMA. In cases of localized overgrowth disorders appearing with megaloccephaly or hemimegaloccephaly due to mutations in the PIK3CA signaling pathway, targeted sequencing can be performed on pathological tissue obtained at autopsy or, alternatively, on cultured amniocytes. However, it should be noted that a negative result does not exclude this diagnosis since this can be a postzygotic variant.⁵⁰

8 | PROGNOSIS

In a study of 17 patients with isolated fetal macrocephaly, the HC was above 2SD in only 11 cases and remained enlarged beyond 18 months in only six cases, suggesting the limited diagnostic accuracy of prenatal ultrasound.⁵¹ The degree of macrocephaly is important, with mild macrocephaly ≤ 2.5 SD carrying a good prognosis, especially when one of the parents has macrocephaly and normal development. In cases of confirmed mild macrocephaly postnatally, that is, with fetal HC between +2 and +3 SD (corresponding to the 99.7th centile)²⁴ normal postnatal development can be expected in most cases with up to 60% found to have a recurrence in future pregnancies.^{5,24} On the other hand, children with a HC exceeding 3SD typically present with neurogenetic disorders characterized by intellectual disability (ID), autism spectrum disorders (ASD), and frequent comorbidities.⁵² In non-isolated cases, the prognosis depends on the underlying cause and associated anomalies.

9 | IN-UTERO TREATMENT

Intrauterine ventriculoamniotic shunting is an experimental procedure for severe ventriculomegaly. Attempts at such therapy in the 1980s were generally abandoned because the majority of survivors had severe neurodevelopmental delay.⁵³ New interest in this intervention has recently risen, and its feasibility has now been demonstrated in several studies.^{54,55} Prenatal endoscopic ventriculostomy

of the 3rd ventricle (ETV) for congenital aqueductal stenosis has been successful in a large animal model.⁵⁶ Nonetheless, the beneficial effect of these fetal interventions on neurodevelopmental outcomes remains to be proven.

10 | OBSTETRIC MANAGEMENT

There is an association between a HC >95th centile and higher risk of unplanned cesarean delivery, instrumental delivery⁵⁷ and obstetric anal sphincter injury (OASI).⁵⁸ Therefore, in fetuses with macrocephaly with a HC >95th centile at term when pelvic-cephalic disproportion is likely during labor, a planned cesarean delivery should be considered. In cases in which macrocephaly is the result of hydrocephalus and the prognosis is grim, pre-labor cephalocentesis can be considered to facilitate vaginal delivery. In most cases, this will result in major morbidity and/or fetal death.⁵⁹

11 | CONCLUSION

In conclusion, fetal macrocephaly can be non-megaloccephalic or megaloccephalic, with numerous causes for each type. The diagnosis of fetal macrocephaly should be approached carefully due to limitations in the accuracy of HC measurement and inconsistencies between prenatal and neonatal HC curves. The vast majority of megaloccephaly cases with a HC below 2.5 SD are familial and benign; however, HC greater than 2.5 SD is more suggestive of a syndromic cause, particularly when parental HC are normal. A detailed anatomical survey (assessing viscera, limbs, skin), neurosonography and MRI are warranted to aid in diagnosis and a multidisciplinary approach is key. In cases of post-zygotic variant genetic investigations should include a biopsy of affected tissue.

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

Not applicable to this review.

ORCID

Shiri Shinar  <https://orcid.org/0000-0001-6352-4846>

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